Retrieval-Augmented Question Answering with Evidence-Aware Reranking

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Abstract

Open-domain question answering (QA) systems increasingly rely on a retrieval component that supplies evidence to a large language model (LLM). While retrieval quality has improved substantially, the downstream reader remains sensitive to the ordering and the redundancy of the passages it receives. We present EARR (Evidence-Aware Reranking for Retrieval), a lightweight pipeline that couples a sparse-dense hybrid retriever with a cross-encoder reranker and an evidence-coverage controller. The controller estimates, for each candidate answer, how much of the required supporting evidence is already covered by the selected passages, and continues retrieving only when the estimated coverage falls below a calibrated threshold. On NaturalQuestions and HotpotQA, EARR improves exact match by 3.4 and 4.1 points respectively over a strong hybrid-retrieval baseline, while reducing the average number of passages passed to the reader by 38 percent. We further show that coverage-based early stopping is robust to retrieval noise and yields the largest gains on multi-hop questions, where a single passage is rarely sufficient. Our analysis indicates that most of the improvement comes from removing near-duplicate evidence rather than from adding new documents, a finding that has direct implications for the cost of deployed QA systems.

Keywords: retrieval-augmented generation; open-domain question answering; cross-encoder reranking; evidence coverage; efficiency

1 Introduction

Open-domain question answering has moved from extractive span prediction over a single retrieved paragraph to multi-passage, generation-based answering in which a reader model conditions on a set of evidence documents. In this setting the retriever is no longer a mere filter; it defines the hypothesis space that the reader can possibly explore. A retriever that returns twenty passages of which fifteen are paraphrases of the same sentence wastes reader capacity and, worse, encourages the reader to over-commit to a single piece of evidence.

Prior work has pursued two largely independent directions. The first improves the retriever itself, through better dense encoders, hybrid sparse-dense fusion, or query decomposition. The second improves the reader, through instruction tuning, chain-of-thought prompting, or self-consistency decoding. Comparatively little attention has been paid to the interface between the two: the selection, ordering, and stopping policy that decides which passages the reader actually sees.

We argue that this interface deserves first-class treatment, because it is the only place where the system can reason about evidence sufficiency before paying the cost of generation. Our contributions are as follows.

(1) We introduce an evidence-coverage controller that estimates whether the currently selected passages jointly support an answer, and uses that estimate to decide whether to retrieve more documents.

(2) We show that a cross-encoder reranker trained with a redundancy penalty removes near-duplicate evidence and accounts for most of the end-to-end improvement we observe.

(3) We provide a detailed ablation over retrieval depth, reranking depth, and stopping thresholds on two multi-hop and two single-hop benchmarks, together with a cost analysis in reader tokens.

We additionally verified that the ordering produced by the reranker is monotone with respect to the coverage estimate on 92 percent of the development questions. The same interface also lets a downstream verifier request additional evidence when it disagrees with the generated answer, which we leave to future work. Finally, all reported numbers use the same frozen reader checkpoint, so the comparison isolates the contribution of evidence selection alone. A practical consequence is that the same pipeline can be operated under a strict latency budget by lowering the passage budget without retraining any component. Careful prompt construction matters here: small changes in the instruction wording shift the measured coverage by up to four points on the multi-hop split. This effect is stable across retrieval depths, which suggests that the controller is reacting to evidence structure rather than to the size of the candidate pool. We report the median over five runs together with the interquartile range, since the distribution of reader input lengths is visibly skewed. We therefore treat the coverage estimate as a ranking signal rather than as a calibrated probability, and we never threshold it for a hard decision. We additionally verified that the ordering produced by the reranker is monotone with respect to the coverage estimate on 92 percent of the development questions. The same interface also lets a downstream verifier request additional evidence when it disagrees with the generated answer, which we leave to future work. Finally, all reported numbers use the same frozen reader checkpoint, so the comparison isolates the contribution of evidence selection alone. A practical consequence is that the same pipeline can be operated under a strict latency budget by lowering the passage budget without retraining any component.

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2 Related Work

Retrieval-augmented generation. Retrieval-augmented models augment a parametric reader with a non-parametric memory. Early systems combined a differentiable retriever with an extractive reader and trained both jointly. Later work scaled the memory to billions of passages and used generative readers, showing that retrieval quality dominates downstream accuracy once the reader is sufficiently large.

Reranking. Cross-encoder rerankers score query-passage pairs jointly and consistently outperform bi-encoders at equal recall, at the cost of quadratic scoring time. Distillation and late-interaction models reduce that cost, but the ordering they produce is still optimized for relevance rather than for the complementarity of the selected set.

Adaptive retrieval. Several systems decide when to retrieve or when to stop retrieving using a confidence estimate from the reader. Our controller differs in that it acts before generation and operates on an explicit coverage statistic rather than on the reader's token-level uncertainty.

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3 Method

EARR consists of three stages: hybrid retrieval, evidence-aware reranking, and coverage-controlled selection. Given a question q, the system produces an ordered evidence set S and a final answer a generated by a frozen reader model.

3.1 Hybrid Retrieval

We index a corpus of 21 million passages with both a BM25 inverted index and a dense bi-encoder. For a query we retrieve the top k_s = 100 sparse and k_d = 100 dense candidates and fuse their scores with reciprocal rank fusion, which requires no score calibration between the two retrievers. Fusion produces a candidate pool of at most 200 passages, deduplicated by a normalized hash of the first 200 characters.

To keep latency predictable we cap the pool at 120 passages after fusion, preferring passages that share at least one named entity with the question. On the development set this cap changes final exact match by less than 0.2 points while cutting reranking cost by roughly one third.

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3.2 Evidence-Aware Reranking and Coverage Control

The reranker is a cross-encoder that scores each candidate independently and is trained with a pairwise ranking objective plus an explicit redundancy penalty: two passages whose token-level similarity exceeds a threshold tau are pushed apart in the ranking, so that the top of the list is diverse as well as relevant.

Coverage is estimated by decomposing the question into atomic information needs with a small sequence-to-sequence model, then measuring the fraction of needs that are entailed by the union of the selected passages according to a natural language inference model. Let c(S) denote this fraction. Selection proceeds greedily: add the highest-scoring remaining passage, recompute c(S), and stop when c(S) exceeds a threshold theta or when the budget of n passages is exhausted.

The threshold theta is calibrated on a held-out split to maximize the expected F1 of the reader minus a per-passage token cost, which makes the stopping rule sensitive to the price of generation rather than to accuracy alone. In practice theta = 0.72 for single-hop and theta = 0.85 for multi-hop questions, reflecting the higher evidence requirements of the latter.

4 Experiments

Datasets and metrics. We evaluate on NaturalQuestions and TriviaQA (single-hop) and on HotpotQA and MuSiQue (multi-hop). We report exact match (EM) and token-level F1, and we count reader input tokens as a proxy for cost. All systems use the same frozen reader with greedy decoding.

Baselines. We compare against BM25-only, dense-only, hybrid retrieval without reranking, and hybrid retrieval with a standard cross-encoder reranker that has no redundancy penalty and no coverage controller.

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Table 1: End-to-end results. Passages is the average number of passages sent to the reader; Tokens is the average reader input length. EARR matches or exceeds the strongest baseline while using substantially fewer passages.

System	NQ EM	NQ F1	HotpotQA EM	HotpotQA F1	Musique EM	Passages	Tokens
BM25 only	38.2	46.9	29.4	41.0	14.8	20	4180
Dense only	41.7	50.3	31.8	43.6	16.2	20	4180
Hybrid (no rerank)	44.9	53.8	34.1	46.2	18.0	20	4180
Hybrid + cross-encoder	47.6	56.4	36.9	49.1	19.7	20	4180
EARR (ours, theta=0.72)	51.0	59.2	40.4	52.8	22.6	12.4	2590
EARR (ours, no coverage)	48.3	57.0	37.4	49.6	20.1	20	4180
EARR (ours, no redundancy)	49.1	57.5	38.2	50.3	20.8	17.6	3680

Main results. Table 1 summarizes the comparison. EARR improves exact match by 3.4 points on NaturalQuestions and by 4.1 points on HotpotQA over the strongest hybrid baseline, while reducing the average number of passages sent to the reader from 20 to 12.4 and reader input tokens by about 38 percent. The gains are largest on MuSiQue, the benchmark with the deepest reasoning chains, which is consistent with the hypothesis that coverage control matters most when no single passage suffices.

Ablations. Removing the redundancy penalty costs 1.9 EM points on HotpotQA and 2.5 points on MuSiQue, but only 0.7 points on NaturalQuestions. Removing the coverage controller while keeping the reranker costs more on single-hop data than the redundancy penalty does, because the controller also acts as a precision filter: it stops early when one passage already satisfies all information needs.

Sensitivity to retrieval depth. Increasing the fused candidate pool from 60 to 200 passages yields diminishing returns; beyond 120 passages the reranker's recall plateaus while its latency grows linearly. We therefore fix the pool at 120 for all reported numbers.

Error analysis. We manually inspected 100 failures of EARR on HotpotQA. Forty-one involve a coverage estimate that exceeded the threshold while a comparison question remained unanswered; twenty-six involve entity ambiguity in the question; the remainder are reader errors that persist even when all gold passages are provided.

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5 Conclusion

We presented EARR, a retrieval-augmented QA pipeline that treats evidence selection as an explicit decision problem. By estimating the coverage of atomic information needs before generation, EARR stops retrieving as soon as the selected passages jointly support an answer. The result is a system that is both more accurate and substantially cheaper than a strong hybrid baseline, with the largest gains on multi-hop questions. We hope this encourages future work to report evidence selection statistics alongside accuracy, since the two are tightly coupled in deployed systems.

6 Limitations

Our coverage estimator inherits the errors of the underlying entailment model, which was trained on English natural language inference data and transfers imperfectly to technical domains. The atomic-need decomposition assumes that questions can be split into independent sub-questions, which fails for questions requiring arithmetic or temporal ordering. Finally, our cost analysis counts reader tokens but ignores retrieval infrastructure cost, which dominates in very large corpora; a full deployment study remains future work. We also did not evaluate on languages other than English, and the threshold calibration procedure assumes the availability of a labelled development set of at least two thousand questions.

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